

Statistics primer & exponential families

“Algebraic & Geometric Methods in Statistics”

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Created for Math/Stat 561

Feb 3, 2026.

Objective

- Review some statistics fundamentals;
- understand the setup of exponential families.

Material:

Sourced from chapters 5 (“Statistics primer”) and 6 (“exponential families”) of the textbook. Other resources provided in subsequent links.

Parametric models: self-review

- What is a **parametric** statistical model?
- What is an **implicit** statistical model?
- What does it mean for random variables $X_1, \dots, X_n$ to be *iid* (**independent and identically distributed**)?
- What does it mean for random variables to be **exchangable**?
- What is an *iid* sample?

Task:

Look up, write down, and adopt these definitions.

See Sections 5.1. and 5.2. of the textbook ([handout](#)).

The running example

Example 5.3.2.: Binomial random variable: $r + 1$ states, $0, \dots, r$.

- The **model** consists of all distributions of the form

$$\{(\pi^r, r\pi^{r-1}(1-\pi), \dots, (1-\pi)^r) : \pi \in [0, 1]\}.$$

In other words, the model is the set $\{p_\pi\} \subset \mathbb{R}^{r+1}$ where each p_π has the above form.

- **Data** collected from an ***iid* sample** of size n : $X^{(1)}, \dots, X^{(n)}$, from an underlying distribution p_{π_0} .
- π_0 is the unknown but fixed **parameter** we would like to **estimate** using the data.

Statistics vs. parameters

Parameter [Definition 5.3.1.]

Let $\mathcal{M}_\theta$ be a parametric statistical model with parameter space Θ . A **parameter** of a statistical model is a function $s : \Theta \rightarrow \mathbb{R}$.

Statistic [Definition 5.1.5.]

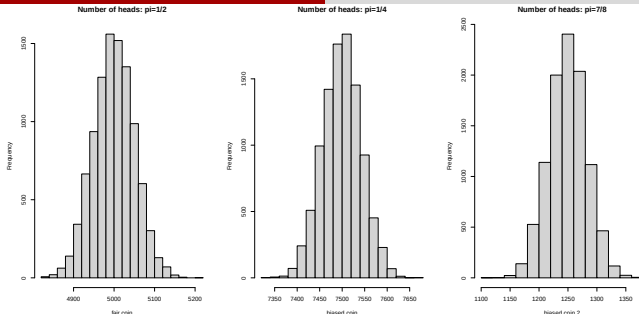
A **statistic** is a function from the state space to some other set.

- A statistic $T(X)$ is **sufficient** for the model if $P(X = x | T(X) = t, \theta) = P(X = x | T(X) = t)$.

Estimator [Definition 5.3.1.]

An **estimator** $\hat{\theta}$ is a function from the data space D to $\mathbb{R}$.

- An estimator is **consistent** if $\hat{\theta} \rightarrow_P \theta$. $\leftarrow$ it converges to the true parameter as the sample size $\rightarrow \infty$.
- There are many ways to compute an estimator.



Three simulations, **one** parametric model, **one** unknown parameter

→ The parametric model: $\text{Bin}(10000, \pi)$

- Histogram 1: data simulated with $\pi = 1/2$.
- Histogram 2: data simulated with $\pi = 1/4$.
- Histogram 3: data simulated with $\pi = 7/8$.

What is the parameter estimation problem on this example?

Write it out.

(What is X , θ or π , r , n , a statistic, an estimator of a parameter?)

The parameter estimation problem

- There are many ways to compute estimators.
 - See Math 563, for starters; Method of moments, for example
 - **READ** Examples in the book re: binomial r.v.: 5.3.2, 5.3.4, 5.3.6.

Maximum likelihood estimation [Defn. 5.3.5.]

Let D be data from some model w/ parameter space Θ . **Likelihood function**:

$$L(\theta|D) := p_{\theta}(D) \text{ or } L(\theta|D) := f_{\theta}(D).$$

- L is a function of the parameter(s)! Data is *fixed* in the likelihood function.

The maximum likelihood estimate (**MLE**) $\hat{\theta}$ is the maximizer of the likelihood function:

$$\hat{\theta} = \arg \max_{\theta \in \Theta} L(\theta|D).$$

- MLE = the particular value $\hat{\theta}$ of the parameter that makes D most likely to have been observed under the model.

Let's look at the Binomial example again.

- The data $D = X^{(1)}, \dots, X^{(n)}$ is summarized by a vector of **counts** $u = (u_0, \dots, u_r)$, where $u_i = |\{j : X(j) = i\}|$.
- In the case of discrete data, this likelihood function is thus only a function of the vector of counts u :

$$L(\theta|D) = \prod_j p_\theta(j)^{u_j}.$$

* It is common to study the **log-likelihood function** $\ell(\theta|D) = \log L(\theta|D)$.

Binomial likelihood

Go over **Example 5.3.6**.

- What is the likelihood function?
- What is the MLE?

MLE for $\mathcal{M}_{1 \perp\!\!\!\perp 2}$.

Go over Proposition 5.3.8. and proof.

Other resources

- Check out this [lovely tutorial on MLE](#) by Prof. Andrew Moore.
- Larry Wasserman's intermediate statistics notes on likelihood and sufficiency: read [this](#) and [this](#).

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